

Monopoly

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A companion to the lecture slides. Each slide that hides a step is reproduced here and worked through line by line.

Before we begin

Required readings

Required readings

The core text is N. Gregory Mankiw, *Principles of Microeconomics*.

- **Chapter 16** — Monopoly.

Read it alongside these slides.

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Slide 2

Working through it

Why this chapter. Chapter 16 covers monopoly: the single seller, how it arises, why marginal revenue lies below price, the output rule, the welfare comparison with competition, price discrimination and public policy.

What the slides add. Marginal revenue is built as a *column of revenue differences* rather than a derivative, and the gap between MR and P is decomposed explicitly into an output

effect and a price effect — which is the single idea the whole session rests on. The natural-monopoly regulation in Part 6 is done *numerically*, so the impossibility of pricing at marginal cost is demonstrated rather than asserted.

What is not here. Monopolistic competition and oligopoly are not covered here. Each is a model in its own right and each has its own treatment.

How to use it. Read chapter 16 before the session. The numerical regulation in Part 6 makes most sense once the monopolist's procedure is automatic.

Where we are going

Where we are going

1. **One seller** — what a monopoly is, and the barriers that keep it alone.
2. **The monopolist's revenue** — why selling more means earning less per unit:
 $MR < P$.
3. **Profit maximization** — the quantity, the price, the profit — and the loss.
4. **Monopoly vs. competition** — the trades that die, and what they were worth.
5. **Price discrimination** — one good, many prices.
6. **What can be done?** — regulating a natural monopoly, with numbers.
7. **Extensions** — the Austrian critique of the whole apparatus.

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Slide 3

Working through it

The structure. Seven parts. Six on monopoly, one critique.

Parts 1–3 build the monopolist. What it is and what protects it (1); why its revenue behaves differently from a price taker's (2); and how it picks a quantity and then a price (3).

Parts 4–6 judge it. What its behavior costs society, in dollars (4); what changes if it can charge different buyers differently (5); and what a regulator can actually achieve against a natural monopoly (6).

Part 7 steps outside the model and asks whether the whole snapshot approach measures the right thing.

The single thread. Everything follows from one fact: a monopolist must cut the price on *all* units to sell one more.

1. Part 1: One Seller

The opposite pole

The opposite pole

Monopoly. A firm that is the sole seller of a product with no close substitutes.

A competitive firm is a **price taker**: the market hands it a price, and its only choice is quantity.

A monopolist is a **price maker**: with no rivals, it chooses the price itself — constrained only by what buyers will take.

Market power is the ability to raise price without losing every customer.

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Slide 5

Working through it

The definition. A **monopoly** is a firm that is the sole seller of a product with no close substitutes.

Both halves of that definition matter. “Sole seller” is not enough: the only steakhouse in town competes with every other restaurant and with cooking at home. It is the absence of *close substitutes* that confers power, and Active Learning 1 turns on exactly that discrimination.

The contrast with the competitive firm.

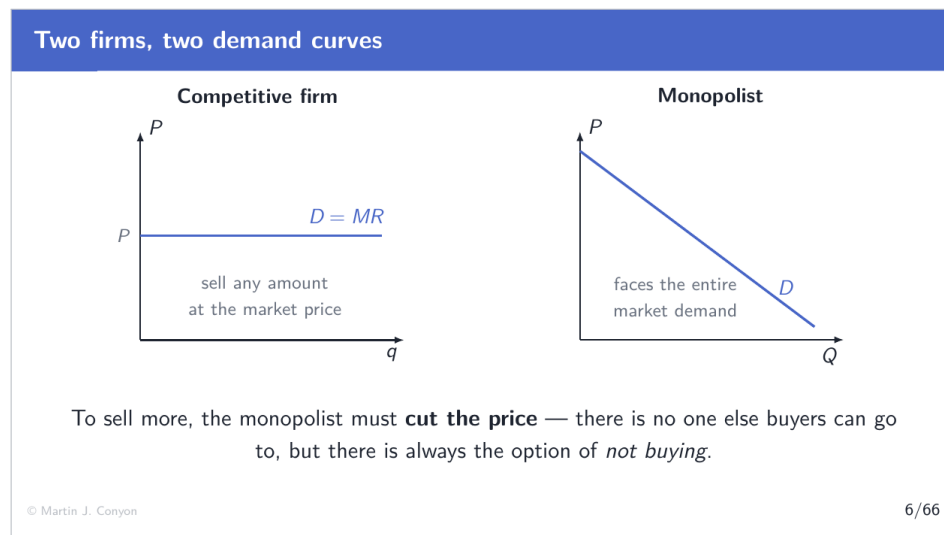
- A competitive firm is a **price taker**. The market hands it a price; its only choice is

quantity.

- A monopolist is a **price maker**. With no rivals it chooses the price, constrained only by what buyers will take.

Defining market power carefully. It is the ability to raise the price *without losing every customer*. Not without losing any — a monopolist that raises its price does lose some buyers, because they can always buy less or buy nothing. The distinction between losing some and losing all is the entire difference between the two structures, and it is what makes the monopolist's demand curve slope rather than lie flat.

Two firms, two demand curves



Slide 6

Working through it

Two pictures, and every result in the session comes from the difference between them.

Left: the competitive firm. A horizontal line at the market price. A penny above it and every buyer moves to an identical rival, so the firm sells nothing. It can sell as much as it likes at the market price and nothing at all above it.

Right: the monopolist. The *entire market demand curve*, sloping down. There is no rival to switch to — but buyers always retain the option of buying less, or not buying.

The consequence, and it is the hinge of the session. To sell one more unit, the monopolist must **cut the price**. The competitive firm never does: it sells its extra unit at the going price, leaving everything else untouched.

Why this is worth dwelling on before any arithmetic. It is easy to imagine that a monopolist simply charges what it wants. It cannot. The demand curve is a constraint just as real as a competitor would be, and Part 2 shows that it is a surprisingly expensive one.

Barriers to entry

Barriers to entry

Why doesn't profit attract rivals, the way it did in a competitive market? Something blocks the door.

1. **A key resource in one firm's hands** — e.g., control of the ore supply (the aluminum trust broken up in *U.S. v. Alcoa*, 1945). Rare today.
2. **A government grant** — patents, copyrights, licenses: the law itself keeps rivals out.
3. **Natural monopoly** — one firm can serve the whole market more cheaply than two could.

No barrier, no monopoly: free entry always erodes the power.

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Slide 7

Working through it

The question this frame answers. In a competitive market, profit attracted entrants and was competed away. Why not here? Because something blocks the door.

The three kinds of barrier.

1. **A key resource in one firm's hands.** Control of the essential input. *U.S. v. Alcoa* (1945) broke up an aluminium monopoly built on control of bauxite. This barrier is rare today, because most resources are traded internationally.
2. **A government grant.** Patents, copyrights, licences: the law itself keeps rivals out. This is the commonest source of durable monopoly, and Part 7 makes something of that.

3. **Natural monopoly.** One firm can serve the whole market more cheaply than two could, so the cost structure does the excluding.

The rule underneath all three. No barrier, no monopoly. Free entry always erodes market power, so any monopoly that persists must be explained by something keeping entrants out. When you meet a firm with apparent market power, the productive question is not “is it big?” but “what is the barrier?”

Barriers by law: the patent trade-off

Barriers by law: the patent trade-off

A patent hands an inventor the exclusive right to sell — typically for 20 years.

- **The cost:** for 20 years, one seller, monopoly pricing — everything measured here.
- **The benefit:** the prospect of those profits is why firms sink fortunes into inventing at all.

Society accepts a *temporary* monopoly as the price of the invention existing.

The monopoly is not a bug in patent law; it is the payment.

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Slide 8

Working through it

What a patent does. It hands an inventor the exclusive legal right to sell, typically for twenty years. Society deliberately creates a monopoly.

The cost. Twenty years of exactly what Parts 2 to 4 measure: one seller, a price above marginal cost, output held short, and a deadweight loss triangle. This is not a side effect — it is the whole point of the grant, from the inventor’s side.

The benefit. An idea, once published, is copyable at near-zero cost. If rivals could copy it immediately, nobody would recover the cost of inventing, and much invention would simply not happen. The prospect of monopoly profit is what funds the research.

The trade-off stated honestly. Society accepts a *temporary* monopoly as the price of the

invention existing at all. The twenty-year term is an attempt to balance the two — long enough to reward the inventor, short enough to limit the damage.

The sentence to carry. The monopoly is not a bug in patent law; it is the payment. And Part 4's patent-expiry frame shows precisely what society gets back when the payment stops.

Natural monopoly

Natural monopoly

Natural monopoly. An industry in which a single firm can supply the entire market at lower average cost than two or more firms could.

Our example: **electricity distribution** in a small city — one grid of poles and wires, cost $TC = 160 + 10Q$ (\$ thousands per month, Q = thousands of homes).

$$ATC = \frac{160}{Q} + 10 \quad \text{— falls at every } Q.$$

The fixed cost is the wires; adding one more home costs only \$10. Spreading \$160 over more homes never stops helping.

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Working through it

The definition. A **natural monopoly** is an industry in which a single firm can supply the entire market at lower average cost than two or more firms could.

The example, used again in Part 6. Electricity distribution in a small city: one grid of poles and wires. With Q in thousands of homes and money in thousands of dollars per month,

$$TC = 160 + 10Q \quad \implies \quad ATC = \frac{160}{Q} + 10$$

Reading the cost function. The **160** is the wires — a fixed cost paid once, whether one home is connected or ten thousand. The **10** is the marginal cost of connecting one more home, which is small.

Why ATC falls at every quantity. $160/Q$ shrinks as Q grows and never turns, and the other term is constant. So spreading the fixed cost over more homes *never* stops helping — there is no efficient scale, no bottom to the average cost curve within the relevant range.

Why that creates a monopoly without anybody intending one. Bigger is always cheaper here, so the largest firm can always undercut a smaller rival. The next frame puts numbers on it.

Why a second grid would be waste

Why a second grid would be waste

Serve the city's 8,000 homes with $ATC = 160/Q + 10$:

	one firm	two firms
homes per firm	$Q = 8$	$Q = 4$ each
average cost per home	$160/8 + 10 = \text{\$30}$	$160/4 + 10 = \text{\$50}$

A rival would have to **build a second grid** to compete — duplicating the \$160 fixed cost to split the same homes.

Here the barrier to entry is the cost curve itself.

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Slide 10

Working through it

The comparison. Serve a city's 8,000 homes, using $ATC = 160/Q + 10$.

- **One firm:** $Q = 8$, so $ATC = 160/8 + 10 = \text{\$30}$ a home.
- **Two firms:** $Q = 4$ each, so $ATC = 160/4 + 10 = \text{\$50}$ a home.

Why the second arrangement is more expensive. The \$160 fixed cost is *duplicated* — two sets of poles and wires — while the same 8,000 homes are split between them. Each firm's fixed cost is spread over half as many customers.

The consequence for policy, and it is the uncomfortable part. Competition here does not discipline prices. It raises the real cost of serving every household by \$20. Encouraging entry would make the city poorer, which is the opposite of the usual prescription.

So the barrier is the cost curve itself. No law is needed and no resource is hoarded. A potential entrant simply cannot match the incumbent's average cost without capturing most of the market first — which it cannot do while the incumbent is cheaper.

Where this leads. Part 6 asks what a regulator can do about it, and finds that every available answer costs something.

2. Part 2: The Monopolist's Revenue

Write the equation

Write the equation

Our running firm: the **sole maker of a specialized industrial sensor**, protected by a patent. Market demand:

$$P = 100 - 5Q \quad (Q \text{ in thousands per month, } P \text{ in dollars})$$

- Choose $Q = 4$, and the demand curve answers: $P = 100 - 5 \times 4 = 80$.
- Choose $P = 60$, and it answers back: $Q = 8$.

Price maker, yes — but the demand curve is the menu. One choice, not two.

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Working through it

The firm. Sole maker of a specialized industrial sensor, protected by a patent. Market demand:

$$P = 100 - 5Q$$

with Q in thousands per month and P in dollars.

What “price maker” actually means, and it is narrower than it sounds. The firm may pick any point it likes *on this line*. It may not pick a point off it.

- Choose $Q = 4$, and the demand curve answers: $P = 100 - 20 = 80$.

- Choose $P = 60$, and it answers back: $Q = 8$.

One choice, not two. Naming a price determines the quantity; naming a quantity determines the price. A monopolist that announces both a high price *and* a large quantity is announcing something buyers will not deliver.

The correction to the popular picture. A monopolist does not charge whatever it likes. It faces a constraint — the willingness of buyers to buy — and the whole of Part 2 is about how expensive that constraint turns out to be at the margin.

Plug the numbers in

Plug the numbers in			
Q	$P = 100 - 5Q$	$TR = P \times Q$	$MR = \Delta TR$
0	100	$100 \times 0 = 0$	—
1	95	$95 \times 1 = 95$	95
2	90	$90 \times 2 = 180$	85
3	85	$85 \times 3 = 255$	75
4	80	$80 \times 4 = 320$	65
5	75	$75 \times 5 = 375$	55
6	70	$70 \times 6 = 420$	45
7	65	$65 \times 7 = 455$	35
8	60	$60 \times 8 = 480$	25
9	55	$55 \times 9 = 495$	15
10	50	$50 \times 10 = 500$	5
11	45	$45 \times 11 = 495$	−5

MR runs below P in every row after the first.

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Slide 15

Working through it

Building the table. For each quantity, read the price off demand, multiply to get total revenue, and difference to get marginal revenue.

$$Q = 7 : P = 65, TR = 455 \quad Q = 8 : P = 60, TR = 480$$

The puzzle the table sets up. The eighth thousand sensors *sold for \$60*. But total revenue rose by only $480 - 455 = \$25$.

Why that is genuinely strange at first sight. For a competitive firm the two numbers are the

same: sell a unit at \$48, and revenue rises by \$48. Here the unit sold for \$60 and revenue rose by \$25, so \$35 has gone somewhere.

What to check before reading on. The arithmetic is not in doubt — both revenue figures come straight from $P \times Q$ with prices read off the demand curve. So the \$35 is not an error. Something real is consuming it, and the next frame identifies it exactly.

The general pattern in the column. Every marginal revenue entry falls short of its own price, and the gap widens as quantity grows. Both facts follow from the same mechanism.

Where did the other \$35 go?

Where did the other \$35 go?

The 8th unit sells at $P = 60$. Why did revenue rise by only 25?

To sell 8 rather than 7, the price must fall from 65 to 60 — **on every unit**:

Output effect: one more unit sold, at \$60	+60
Price effect: \$5 less on each of the 7 units already sold	$7 \times (-5) = -35$
net change in revenue (= MR , the table's row 8)	+25

To sell one more, cut the price on all — so $MR < P$, always.

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Slide 16

Working through it

The decomposition, and it is the central result of the session. To sell 8 rather than 7, the price must fall from \$65 to \$60 — and it falls on *every* unit, not only the new one.

- **Output effect:** one more unit sold, at \$60. +60
- **Price effect:** \$5 less on each of the 7 units already being sold. $7 \times (-5) = -35$

Net: +\$25, which is exactly the table's entry.

Why the price effect exists at all. The firm sells at one price. It cannot offer the eighth unit at \$60 while keeping the first seven at \$65 — that would be price discrimination, which requires conditions Part 5 sets out and which this firm does not have.

The general conclusion. $MR < P$ **always**, for any firm facing a downward-sloping demand curve. And the gap is the price effect, which grows with the number of units already being sold.

Why a competitive firm escapes it. Its demand curve is flat, so selling one more requires no price cut at all. The price effect is zero, and $MR = P$. The two structures differ in exactly this one term.

Revenue has a summit

Revenue has a summit

Read the *MR* column again: +5 for the 10th unit, **−5** for the 11th.

- Through $Q = 10$: output effect wins — revenue still climbing.
- From $Q = 11$: **price effect wins** — selling more *shrinks* revenue.

Total revenue peaks at $Q = 10$: $TR = 50 \times 10 = 500$.

Beyond the summit, every extra sale costs money. No firm sells there on purpose.

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Working through it

Reading further down the MR column. The 10th unit adds +\$5. The 11th adds **−\$5** — selling it *reduces* total revenue.

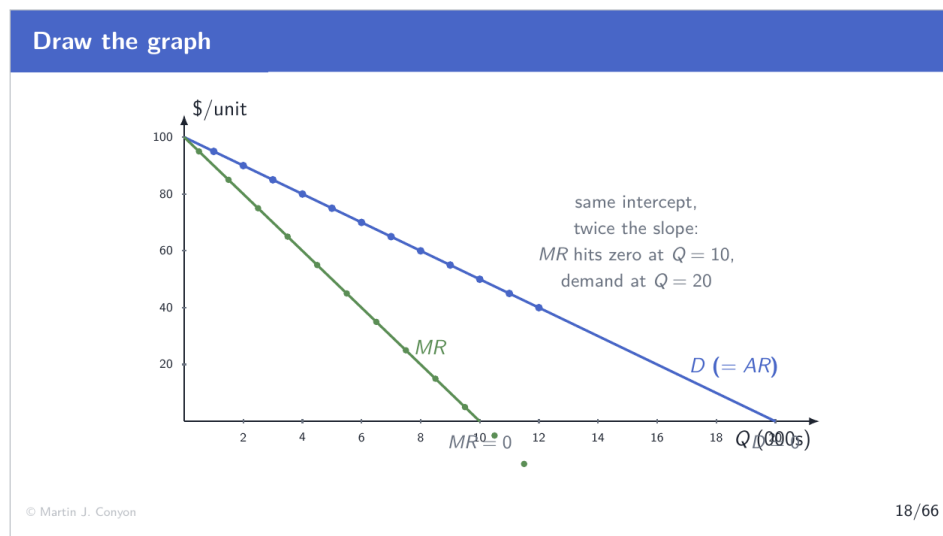
Why marginal revenue turns negative. The two effects are fighting, and their relative sizes change as quantity grows. The output effect is roughly the price, which is falling. The price effect is the price cut times the number of units already sold, which is rising. Eventually the second overtakes the first.

Where the summit is. Through $Q = 10$ the output effect wins and revenue climbs; from $Q = 11$ the price effect wins and revenue falls. So total revenue peaks at $Q = 10$, where $TR = 50 \times 10 = 500$.

The general rule. Total revenue peaks exactly where marginal revenue changes sign — which is what Active Learning 2 asks you to verify on different numbers.

One conclusion available immediately, before costs are mentioned. No firm would ever choose to sell beyond the summit, because doing so lowers revenue *and* raises cost. So the monopolist's output is certainly below 10 thousand. Once costs enter in Part 3, it turns out to be well below.

Draw the graph



Slide 18

Working through it

Two lines, and the relationship between them is worth memorizing.

Demand: $P = 100 - 5Q$. It starts at 100 and reaches zero at $Q = 20$.

Marginal revenue: $MR = 100 - 10Q$. **Same intercept, twice the slope.** It starts at the same 100 and reaches zero at $Q = 10$ — exactly half way.

Why that relationship holds for any straight-line demand curve. If $P = a - bQ$ then $TR =$

$aQ - bQ^2$, and the increments of that expression fall at twice the rate of P itself. So the “same intercept, twice the slope” rule is general, and it makes marginal revenue easy to draw without computing a table.

The dots. These are the table’s differences, plotted at half-integer positions — the marginal revenue of the 8th unit is the change between $Q = 7$ and $Q = 8$, so it belongs at $Q = 7.5$. They lie exactly on the line, which is the check that table and picture are the same object.

Where MR hits zero. At $Q = 10$, the revenue summit from the previous frame. The picture and the table agree, as they must.

Two revenue rules

Two revenue rules

From the table, for *any* firm:

$$AR = \frac{TR}{Q} = \frac{P \times Q}{Q} = P \quad \text{— average revenue is the price.}$$

	competitive firm	monopolist
demand it faces	horizontal at P	the market's, sloping down
price effect	none	always
marginal revenue	$MR = P$	$MR < P$

Next: set this MR against marginal cost — the same output rule every firm obeys.

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Slide 19

Working through it

The trivial rule. For *any* firm,

$$AR = \frac{TR}{Q} = \frac{P \times Q}{Q} = P$$

Average revenue is the price. This is arithmetic and carries no economic content — it is true of monopolists and price takers alike.

The substantive rule. Whether MR equals P depends entirely on whether a price effect

exists.

- **Competitive firm:** horizontal demand, no price effect, $MR = P$.
- **Monopolist:** sloping demand, price effect on every unit, $MR < P$.

Why the comparison table is worth studying rather than glancing at. Every difference between the two market structures in the rest of the session traces back to this one row. The output rule is the same; the cost curves are the same; only MR differs, and that difference produces the higher price, the withheld output and the deadweight loss.

What comes next. Set this MR against marginal cost — the same $MR = MC$ rule every profit-maximizing firm obeys. The rule is universal; only the curve it is applied to has changed.

3. Part 3: Profit Maximization

The cost side of the firm

The cost side of the firm

The sensor firm's total cost, in \$ thousands per month:

$$TC = 200 + 20Q$$

- **Fixed cost** $FC = 200$ — the patent license and the clean room, owed at $Q = 0$.
- **Marginal cost** $MC = 20$ — every extra thousand of sensors adds the same 20.
- **Average total cost** $ATC = 200/Q + 20$, falling as the fixed bill spreads:

Q	2	4	5	8	10	16
ATC	120	70	60	45	40	32.5

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Working through it

Why the cost function is kept simple. $TC = 200 + 20Q$ is deliberately linear, so that all the interesting behavior comes from the revenue side. In here the novelty is MR , not MC .

Reading it.

- **Fixed cost** $FC = 200$ — the patent licence and the clean room, owed at $Q = 0$.
- **Marginal cost** $MC = 20$, constant. Every extra thousand sensors adds the same \$20 thousand. There are no diminishing returns here.
- **Average total cost** $ATC = 200/Q + 20$, falling throughout as the fixed bill spreads.

The ATC row to note. At $Q = 8$, $ATC = 200/8 + 20 = 45$. That number is used to draw the profit rectangle two frames from now, and it must be read *at the chosen output* — not at any other quantity.

A useful consequence of constant MC. Because marginal cost does not vary, the $MR = MC$ comparison is unusually clean: a falling MR column meets a flat line, so there is exactly one crossing and no ambiguity about which side is which.

Set MR against MC , unit by unit

Set MR against MC , unit by unit				
Each unit is worth making while it adds more than it costs:				
unit	MR	MC	$MR - MC$	verdict
1	95	20	+75	make it
2	85	20	+65	make it
⋮ margins shrink \$10 a row ⋮				
7	35	20	+15	make it
8	25	20	+5	make it
9	15	20	-5	skip it
10	5	20	-15	skip it
The firm produces $Q_m = 8$ — where MR meets MC . Same rule as every firm: produce where $MR = MC$.				
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Slide 24

Working through it

The comparison. Each unit is worth making while it adds more than it costs. MR falls by \$10 a row; MC is flat at \$20.

- Unit 1: $95 - 20 = +75$. Make it.
- Unit 7: $35 - 20 = +15$. Make it.
- Unit 8: $25 - 20 = +5$. Make it, though the margin is thin.
- Unit 9: $15 - 20 = -5$. Skip it.

So $Q_m = 8$ thousand sensors — where MR meets MC .

Why this is the same rule as before. A competitive firm expands while the next unit adds more than it costs, and so does this one. $MR = MC$ is universal. What has changed is that MR is now a *falling* column rather than a constant equal to the price.

The consequence worth noticing now. The firm stops at 8 even though buyers at unit 9 would pay \$55 for a sensor costing \$20 to make. That trade is worth \$35 to the two parties involved, and it does not happen — because making it would require cutting the price on all 8 thousand units already sold. Part 4 counts those lost trades and prices them.

The price comes off the demand curve

The price comes off the demand curve

$MR = MC$ picks the *quantity*. The *price* is what buyers will pay for it:

$$P_m = 100 - 5 \times 8 = 60$$

Three steps, always in this order:

1. Find Q_m where $MR = MC$ $\rightarrow Q_m = 8$
2. Go **up to the demand curve** at that quantity $\rightarrow P_m = 60$
3. Never price at the $MR = MC$ crossing — that \$20 is a quantity signal, not a price

$$P_m = 60 > MR = MC = 20 \text{ — price above marginal cost.}$$

Working through it

The step that trips people, so it gets its own frame. $MR = MC$ picks the *quantity*. It does not pick the price.

Finding the price. Ask what buyers will pay for 8 thousand sensors — which is what the demand curve is for:

$$P_m = 100 - 5 \times 8 = 60$$

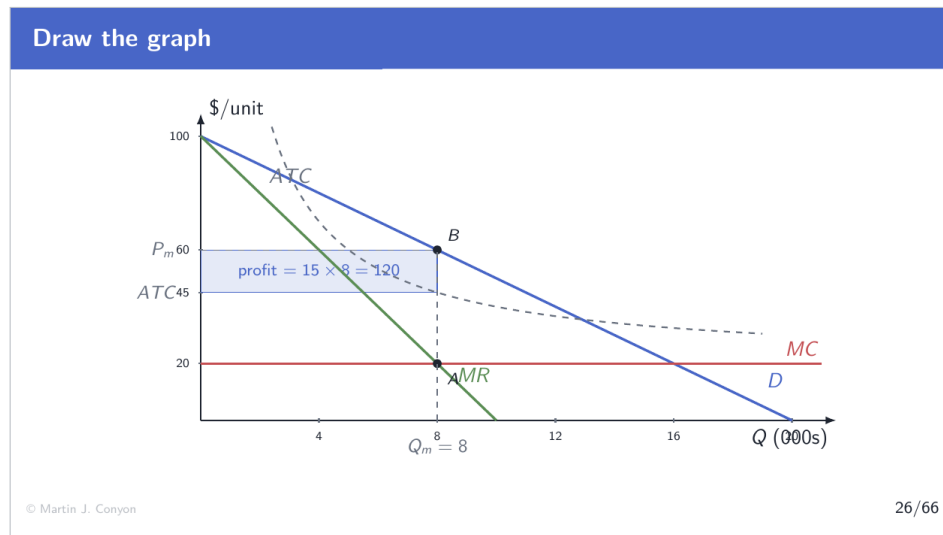
The procedure, always in this order.

1. Find Q_m where $MR = MC$. $\rightarrow Q_m = 8$
2. Go up to the demand curve at that quantity. $\rightarrow P_m = 60$
3. Never price at the $MR = MC$ crossing itself. That \$20 is a *quantity* signal.

The commonest error in the whole topic. Reading the price off the crossing, giving \$20. That would be pricing at marginal cost — which is what a *competitive* market does, and precisely what a monopolist does not.

The result to carry into Part 4. $P_m = 60 > MR = MC = 20$. The price exceeds the marginal cost of the last unit made, and that gap is the source of every welfare conclusion that follows.

Draw the graph



Slide 26

Working through it

Reading the figure in the order the firm decides.

1. The green MR line crosses the flat MC line at $Q = 8$. That fixes the quantity.
2. Climb *vertically* from there to the blue demand curve: \$60. That fixes the price.
3. Drop to the ATC curve at the same quantity: \$45.
4. The gap between price and average cost, \$15, is profit per unit; across 8 thousand units the shaded rectangle is \$120 thousand.

The two points to keep straight. Point A is the $MR = MC$ crossing, at a height of \$20. Point B is the price, at a height of \$60, directly above it. They sit at the same quantity and very different heights, and confusing them is the error the previous frame warned about.

What the picture shows that the algebra does not. The vertical distance between B and the MC line — \$40 — is the wedge between what buyers pay and what the last unit costs. Part 4 turns that distance into the height of the deadweight loss triangle.

The profit, two ways

The profit, two ways

Route 1 — totals:

$$\pi = TR - TC = 60 \times 8 - (200 + 20 \times 8) = 480 - 360 = \mathbf{120}$$

Route 2 — the rectangle:

$$\pi = (P_m - ATC) \times Q_m = (60 - 45) \times 8 = \mathbf{120}$$

Two routes, one answer — if they disagree, an error is hiding.

\$120,000 a month — and **no entrant is coming to erase it**. The patent holds. Monopoly profit can persist.

Working through it

Route 1 — totals.

$$\pi = TR - TC = 60 \times 8 - (200 + 20 \times 8) = 480 - 360 = \mathbf{120}$$

Route 2 — the rectangle.

$$\pi = (P_m - ATC) \times Q_m = (60 - 45) \times 8 = \mathbf{120}$$

Why compute it twice. The two routes share no arithmetic: one subtracts total cost from total revenue, the other multiplies a per-unit margin by a quantity. Agreement is evidence; disagreement would locate an error.

The economically important part. \$120 thousand a month — and **no entrant is coming to erase it**. In a competitive market, profit of this size would summon entry within a year, supply would shift right, and the profit would vanish. Here the patent holds the door shut.

The contrast worth stating explicitly. Competitive profit is temporary by construction; monopoly profit is durable by construction. The difference is not in the firms' behavior — both maximize profit identically — but in whether anything stops others from copying them.

A monopoly can lose money too

A monopoly can lose money too

Same demand, but suppose the license fee doubles the fixed cost: $TC = 400 + 20Q$.

- $MR = MC$ is untouched by fixed cost $\rightarrow$ still $Q_m = 8$, $P_m = 60$.
- But now $ATC = 400/8 + 20 = 70 > 60$: the rectangle flips **above** the price.

$$\pi = (60 - 70) \times 8 = \mathbf{-80} \quad (\text{check: } 480 - 560 = -80)$$

Market power guarantees a captive market — not a profit.

Working through it

Correcting a natural assumption. Market power is often taken to guarantee profit. It does not.

The change. Double the fixed cost — suppose the licence fee doubles: $TC = 400 + 20Q$.

What does not change. $MR = MC$ involves no fixed cost, so the output decision is untouched: still $Q_m = 8$, still $P_m = 60$. A fixed cost cannot change a marginal comparison.

What does change. $ATC = 400/8 + 20 = 70$, which is now *above* the \$60 price. The rectangle flips to sit above the price line:

$$\pi = (60 - 70) \times 8 = -80$$

Check by totals: $480 - 560 = -80$. ✓

The lesson. Market power guarantees a captive market, not a profit. A monopolist with high enough fixed costs loses money and will eventually leave — which is why many patented inventions never reach production at all.

The parallel with the competitive firm. There too, fixed costs never entered the output decision but entirely determined whether the outcome was a profit or a loss. The same separation holds here.

Why a monopoly has no supply curve

Why a monopoly has no supply curve

A competitive firm has a supply curve: *for each price, a quantity* — read off MC .

The monopolist never asks “what will I make at a given price?” It asks: **given the whole demand curve, which point do I pick?**

- Change the demand curve, and the same $MC = 20$ yields a different Q_m and P_m .
- Price and quantity are chosen **jointly**, from MR , MC , and demand together.

No fixed price-to-quantity rule exists — so no supply curve.

Working through it

What a supply curve is. A rule mapping each price to a quantity. A competitive firm has one, read straight off its MC curve, because the firm takes the price as given and answers “how much at that price?”

Why the monopolist has no such rule. It is never asked that question. It asks instead: given the *whole* demand curve, which point do I pick? Price and quantity are chosen *jointly*, from MR , MC and demand together.

The test that makes it concrete. Change the demand curve — make it steeper, say, while leaving $MC = 20$ untouched. The monopolist’s chosen quantity changes *and* its chosen price changes. So the same marginal cost is compatible with many different price–quantity pairs, depending on demand.

Therefore no fixed price-to-quantity relationship exists, and a supply curve is precisely such a relationship. There is nothing to draw.

Why this is more than a technicality. It means the familiar supply-and-demand apparatus cannot be used on a monopolized market. There is a demand curve but no supply curve, and the equilibrium is found by the firm’s optimization rather than by two curves crossing.

4. Part 4: Monopoly vs. Competition

What one trade is worth

What one trade is worth

Rebuild the yardstick in one frame:

- The **demand curve’s height** at any Q is what that unit is *worth* to its buyer — the most they would pay.
- **Marginal cost** is what that unit takes to make.
- If value $>$ cost, making it **creates surplus** — value minus cost — split between buyer and seller.

Sensors: unit 4 is worth $100 - 5 \times 4 = 80$ and costs 20 $\rightarrow$ that trade creates **\$60 of surplus**.

Total surplus = value created minus cost, summed over every unit actually made.

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Working through it

Rebuilding the yardstick, self-contained. To judge the monopolist we need a measure of what a market delivers, so this frame reassembles it in three lines.

- The **height of the demand curve** at any Q is what that unit is *worth* to its buyer — the most they would pay.
- **Marginal cost** is what that unit takes to make.
- If value exceeds cost, making the unit **creates surplus** — value minus cost — split between buyer and seller depending on the price.

A worked unit. Sensor number 4 is worth $100 - 5(4) = \$80$ to its buyer and costs \$20 to make. That single trade creates **\$60** of value that did not exist before.

The measure. Total surplus is value created minus cost, summed over every unit actually made.

Why the price does not appear in that definition. The price determines how the \$60 is *divided* between buyer and seller, not how much there is. That is why a transfer from consumers to a monopolist is not, by itself, a social loss — a point Part 4 makes precisely.

The efficient quantity

The efficient quantity

Make every unit whose value covers its cost — stop where value falls to cost:

$$100 - 5Q = 20 \quad \Rightarrow \quad Q^* = 16$$

A **competitive** sensor industry would land exactly there: price pushed to marginal cost, $P = 20$, $Q = 16$.

Total surplus at $Q = 16$: the triangle under demand above cost,

$$\frac{1}{2} \times (100 - 20) \times 16 = \mathbf{640}.$$

That \$640,000 is the **most this market can deliver** — the benchmark.

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Working through it

The efficient rule. Make every unit whose value covers its cost; stop where value falls to cost.

$$100 - 5Q = 20 \quad \implies \quad Q^* = 16$$

at a price of \$20.

Why a competitive industry would land exactly there. Competition pushes price to marginal cost, and $P = MC$ is the same equation. So the efficient quantity is not an abstract ideal — it is what this market would actually deliver if the patent expired and rivals entered.

Total surplus at Q^ .* The triangle between the demand curve and the \$20 cost line, from 0 out to 16:

$$\frac{1}{2} \times (100 - 20) \times 16 = \mathbf{640}$$

that is, \$640 thousand a month.

Why that number is the benchmark. It is the *most this market can deliver* — there is no arrangement of prices, firms or policies that produces more, because every unit worth making is made and none that is not. Everything in the rest of Part 4 is measured as a shortfall from 640.

The monopolist stops eight units short

The monopolist stops eight units short

The monopolist chose $Q_m = 8$. Look at the units it **refuses to make**:

unit	worth (demand)	costs (MC)	verdict on the trade
9	55	20	worth making — never happens
12	40	20	worth making — never happens
16	20	20	the last one worth making — never happens

Units 9 through 16: all valued at or above cost, all untraded. **Why?** Selling them would force the price down **on every unit** — the price effect again.

The trades that should have happened, and did not — that is the social cost of monopoly.

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Working through it

Where to look. Not at the units the monopolist makes, but at the ones it *refuses* to make — quantities 9 through 16.

Auditing two of them.

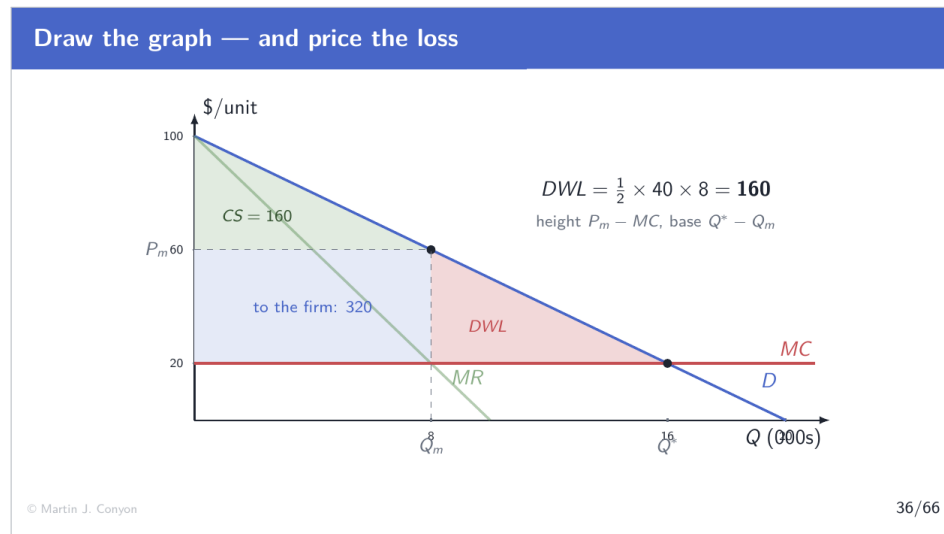
- Unit 9: worth $100 - 45 = \$55$ to its buyer; costs \$20. A trade worth \$35 to the two parties. *Never happens.*
- Unit 12: worth \$40; costs \$20. A trade worth \$20. *Never happens.*

The question this forces. These are trades both sides would gladly make. Why does a profit-maximizing firm refuse them?

The answer, and it is the price effect again. To sell unit 9 the firm must drop its price from \$60 to \$55 — on all 8 thousand units it is already selling profitably. That costs $8 \times \$5 = \40 , against \$55 gained. The firm loses by making the sale, even though the sale itself is worth making from society's point of view.

So the withholding is not spite or inefficiency in production. It is the rational response to facing the whole demand curve. The trades that should have happened and did not are the social cost of monopoly, and the next frame prices them.

Draw the graph — and price the loss



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Working through it

Locating the triangle. It sits between the demand curve and the marginal cost line, spanning the quantities from $Q_m = 8$ to $Q^* = 16$.

Its dimensions.

- **Height:** at $Q = 8$ the gap between value (\$60) and cost (\$20) is \$40 — the wedge the monopoly price creates.
- **Base:** $16 - 8 = 8$ thousand units, the trades not made.
- The gap shrinks steadily to zero at $Q = 16$, where value meets cost, so the region is a triangle.

The computation.

$$DWL = \frac{1}{2} \times 40 \times 8 = \mathbf{160}$$

\$160 thousand a month.

What this number is not. It is not the firm's profit. The profit is \$120 thousand and it goes to *somebody*. The \$160 thousand goes nowhere at all — it is value that would have been

created and is not, because the trades that would have created it never take place. Nobody receives it, which is exactly what makes it a loss rather than a transfer.

Where the consumers' \$640 went

Where the consumers' \$640 went		
Under competition, buyers kept the whole \$640. Under monopoly:		
kept by consumers	160	surplus on units still bought
transferred to the firm	320	higher price on 8 units — moved, not lost
destroyed	160	the 8 units never made — <i>lost to everyone</i>
	640	

The profit is a transfer. The triangle is the loss.

The monopoly price works like a private tax: a wedge between what buyers pay and what production costs — but the “revenue” goes to the firm.

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Working through it

The full accounting, and it must add up. Under competition buyers kept the entire \$640. Under monopoly it divides three ways.

- **Kept by consumers: 160.** The surplus on the 8 thousand units still bought — $\frac{1}{2} \times (100 - 60) \times 8$.
- **Transferred to the firm: 320.** The higher price on those same 8 thousand units — $(60 - 20) \times 8$. This is *moved*, not lost.
- **Destroyed: 160.** The 8 thousand units never made.

$$160 + 320 + 160 = 640. \checkmark$$

The distinction the frame exists to enforce. The **profit is a transfer** — consumers are worse off and the firm's owners are better off by the same amount, so society's total is unchanged by it. The **triangle is a loss** — nobody is better off by any part of it.

Why the two are often confused. Both are consequences of the high price, and both feel like consumers being harmed. But a policy that redistributed the \$320 back would leave the \$160 untouched, and a policy that recovered the \$160 would be a genuine gain to everybody.

The useful analogy. A monopoly price works like a private tax: a wedge between what buyers pay and what production costs. The difference is that the “revenue” goes to the firm rather than to a treasury.

Competition vs. monopoly: the ledger

Competition vs. monopoly: the ledger		
	competition	monopoly
pricing logic	$P = MC$	$MR = MC$, then $P > MC$
price	\$20	\$60
quantity	16,000	8,000
consumer surplus	640	160
producer surplus	0	320
total surplus	640	480
deadweight loss	0	160
long-run profit	erased by entry	guarded by the barrier

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Working through it

The two columns, and the pattern is worth reading down rather than across.

Pricing logic. Competition sets $P = MC$ directly. Monopoly sets $MR = MC$ and then reads the price off demand, which lands it above MC . One rule, two curves.

Outcomes. Price 20 against 60. Quantity 16,000 against 8,000 — the monopolist sells exactly half.

Welfare. Consumer surplus falls from 640 to 160. Producer surplus rises from 0 to 320. Total surplus falls from **640 to 480**, and the **160** difference is the deadweight loss.

The last row, which is easy to skip and is the durable difference. Under competition, long-

run profit is *erased by entry*. Under monopoly it is *guarded by the barrier*. Everything else in this table is about a single moment; that row is about whether the moment persists.

What the table does not say. It does not say monopoly is always bad policy — Part 1 showed patents buy inventions with exactly this loss, and Part 6 shows some monopolies are cheaper than the alternative. It says what the loss is, so that it can be weighed.

The natural experiment: a patent expires

The natural experiment: a patent expires

A brand-name drug's 20-year patent runs out. Generic makers enter the next morning. The market slides from one column of our ledger to the other:

- **Price:** from the monopoly point on demand down toward marginal cost.
- **Quantity:** out from Q_m toward Q^* — patients priced out before are served now.
- **The triangle:** reclaimed — the trades that weren't happening, happen.

Same pill, same factories, same demand — only the barrier changed.

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Why this event is worth a frame. It is as close to a controlled experiment as economics gets. A twenty-year drug patent expires and generic makers enter the next morning. The product, the factories, the disease and the patients are all unchanged. **Only the barrier changed.**

What the model predicts, in order.

- **Price** falls from the monopoly point on the demand curve toward marginal cost.
- **Quantity** rises from Q_m toward Q^* — patients priced out before are now served.
- **The triangle** is reclaimed: trades that were not happening, happen.

What is actually observed. Generic entry routinely cuts prices by large multiples within a year or two, and volumes rise. The direction and rough magnitude match.

Why the experiment is clean. Most economic comparisons are confounded — different countries, different periods, different products. Here everything is held fixed except the legal barrier, so the change in outcomes can be attributed to market structure with unusual confidence.

And the honest reading. It also shows what the patent bought during those twenty years. Both halves of the trade-off are visible in the same event.

5. Part 5: Price Discrimination

One good, many prices

One good, many prices

Price discrimination. Selling the same good to different buyers at different prices, when the cost of serving each buyer is the same.

It takes two ingredients:

1. **Market power** — a price taker cannot charge anyone a premium; rivals undercut it instantly.
2. **No resale** — if bargain buyers can resell to premium buyers, the premium collapses.

Services and licenses discriminate easily (a haircut, a login); goods that ship in boxes do not.

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Working through it

The definition. **Price discrimination** is selling the same good to different buyers at different prices, when the cost of serving each buyer is the same.

The cost clause matters. Charging more for overnight delivery is not price discrimination — it costs more to provide. The practice at issue is charging different prices for something that costs the same to supply.

Two ingredients, and it fails without either.

1. **Market power.** A price taker cannot charge anyone a premium; rivals undercut instantly. So discrimination is only available to firms with some pricing latitude.
2. **No resale.** If bargain buyers can resell to premium buyers, the premium collapses — the discounted units simply flow to the high-price market.

Which is why the practice is concentrated where it is. Services and licences discriminate easily: a haircut cannot be resold, and a software login can be tied to an identity. Goods that ship in boxes are much harder, because boxes travel. That single observation explains most of the pattern of who discriminates and who does not.

The single price leaves money on the table — twice

The single price leaves money on the table — twice

At $P_m = 60$, the sensor monopolist walks past two piles of money:

- **The eager buyers.** Unit 2 was worth \$90 to its buyer — who paid 60 and pocketed 30. The firm would love that 30.
- **The excluded buyers.** Units 9–16 are worth 55 down to 20, all above $MC = 20$ — profitable to serve, but only at prices below 60, which the single price forbids.

One price cannot both harvest the top of the demand curve and serve the bottom.

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Working through it

The setup. At the single monopoly price of \$60, the firm walks past two different piles of money.

Pile one: the eager buyers, at the top of the demand curve. Sensor number 2 was worth \$90 to its buyer, who paid \$60 and kept \$30 of surplus. The firm would very much like that \$30 — and could have had it, if it could charge that buyer separately.

Pile two: the excluded buyers, at the bottom. Units 9 through 16 are worth \$55 down to \$20, all at or above the marginal cost of \$20. Every one of them is profitable to serve — but only at prices below \$60, which the single price forbids.

Why one price cannot capture both. Lowering the price to reach the bottom of the curve gives away margin at the top; raising it to harvest the top excludes even more of the bottom. A single price is one instrument for two jobs.

What discrimination promises. Charge high at the top and low at the bottom, and collect both piles. The next frame pushes that to its logical limit to see how much is at stake.

The extreme case: perfect price discrimination

The extreme case: perfect price discrimination

Suppose the firm knew **every buyer's exact valuation** and charged it:

- Every unit worth ≥ 20 gets sold — out to **Q = 16**, the **efficient** quantity.
- Revenue = the whole area under demand to 16: $\frac{1}{2}(100 + 20) \times 16 = 960$.
- $\pi = 960 - (200 + 320) = \mathbf{440}$ vs. 120 under one price.

	CS	DWL	total surplus
one price	160	160	480
perfect discr.	0	0	640

Efficient — and the firm keeps every dollar of it.

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Working through it

The thought experiment. Suppose the firm knew every buyer's exact valuation and charged precisely it.

What follows.

- Every unit worth at least \$20 is sold, because each buyer pays their own valuation and any valuation above \$20 leaves the firm a margin. Output reaches **Q = 16** — *the efficient quantity*.

- Revenue is the entire area under the demand curve out to 16: $\frac{1}{2}(100 + 20) \times 16 = 960$.
- Profit is $960 - (200 + 320) = 440$, against 120 under a single price.

The result that surprises people. Consumer surplus is **zero** — every buyer pays exactly what the unit was worth to them, so nobody keeps anything. But deadweight loss is *also* zero, and total surplus is the full 640, the same as under perfect competition.

So this outcome is efficient. The right quantity is produced and no trade is lost. The firm simply captures every dollar of the surplus rather than sharing it.

The moral, which is uncomfortable. Efficiency and fairness are genuinely separate questions. Perfect price discrimination scores perfectly on the first and, for buyers, as badly as possible on the second.

Real firms approximate it by *group*

Real firms approximate it by *group*

No firm reads minds. Real discrimination sorts buyers by an **observable trait** that tracks willingness to pay:

- **Student and senior discounts** — ID cards mark the price-sensitive.
- **Peak vs. off-peak fares** — the schedule-bound pay more; matinee crowds pay less.
- **Coupons and promo codes** — charge less only to those who show they will hunt for it.
- **Versioning** — student vs. professional software editions: same code, two prices.

Each is a coarse screen: two or three prices, not a price per buyer.

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Working through it

The practical problem. No firm reads minds. Perfect discrimination requires knowing each buyer's valuation, which nobody does.

The workaround. Sort buyers by an *observable trait* that correlates with willingness to pay, and charge by group.

- **Student and senior discounts.** An ID card marks the price-sensitive.
- **Peak and off-peak fares.** The schedule-bound pay more; the flexible pay less. Note that the airline is not sorting by wealth but by *flexibility*.
- **Coupons and promo codes.** These are self-sorting: only buyers willing to spend time hunting get the discount, and willingness to hunt tracks price sensitivity.
- **Versioning.** A stripped student edition beside a professional one, where the difference in features is often much smaller than the difference in price.

What the schemes have in common. Each finds a *proxy* for valuation that the buyer cannot easily fake, and each is imperfect — some low-value buyers pay the high price and vice versa. Real discrimination captures part of the extra surplus, not all of it.

Is it good or bad? It depends what it replaces

Is it good or bad? It depends what it replaces

Compared with the *single monopoly price*:

- **Profit rises** — always. (Otherwise the firm would not bother.)
- **Output can rise** — discount buyers who were excluded get served: the dead triangle shrinks.
- But surplus **shifts from buyers to the firm** — and a clumsy scheme can lower total surplus.

The only guarantee is higher profit. The welfare verdict is case by case.

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Working through it

The comparison must be specified. Against *perfect competition*, discrimination looks bad. Against a *single monopoly price* — which is the realistic alternative for a firm with market power — the verdict is mixed.

What is certain. **Profit rises, always.** A firm that found discrimination unprofitable would not bother, so observing it tells you profit went up.

What is likely. **Output can rise.** Discount buyers who were excluded at the single price now get served, so the dead triangle shrinks. Active Learning 5 is a case where this effect is the whole story and nobody is made worse off.

What is also true. **Surplus shifts from buyers to the firm.** And a clumsy scheme — one that sorts badly, or adds administrative cost — can lower total surplus even while raising profit.

The honest summary. The only guarantee is higher profit. Whether total surplus rises or falls depends on how much extra output the scheme generates against how much surplus it extracts, and that is a case-by-case empirical question rather than something theory settles.

6. Part 6: What Can Be Done?

Four options, one hard case

Four options, one hard case

1. **Antitrust law** — block mergers, break up dominant firms (Sherman Act 1890, Clayton Act 1914). Works where competition *can* exist.
2. **Regulate the price** — the standard tool where it cannot: the natural monopoly.
3. **Public ownership** — run it as a government enterprise.
4. **Do nothing** — if the cure costs more than the disease.

Option 2 sounds easy — “just cap the price at cost.” **Which cost?** The next frames do it with numbers.

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Working through it

The four instruments.

1. **Antitrust law.** Block mergers, break up dominant firms — the Sherman Act (1890) and Clayton Act (1914). This works where competition *can* exist, so it is the right tool for a monopoly built on acquisition rather than on costs.
2. **Regulate the price.** The standard tool where competition cannot exist — the natural monopoly.
3. **Public ownership.** Run it as a government enterprise.
4. **Do nothing.** If the cure costs more than the disease.

Why breaking up a natural monopoly is the wrong instrument. Part 1 showed two grids cost \$50 a home against one grid's \$30. Antitrust applied here would raise costs, not lower them.

So option 2 is the live one — and the question it begs. “Just cap the price at cost” sounds obvious until you ask **which cost**. Marginal cost and average cost are different numbers for a natural monopoly, and they are different by exactly the amount that decides whether the firm survives. The next three frames work both out on numbers.

The patient: the utility, left alone

The patient: the utility, left alone

From Part 1: $P = 70 - 5Q$, $TC = 160 + 10Q$ (Q = thousands of homes).

$$MC = 10 \quad ATC = \frac{160}{Q} + 10 \quad \Rightarrow \quad \text{MC sits below ATC at every } Q.$$

Unregulated, it behaves like any monopolist (MR falls twice as fast: $70 - 10Q$):

$$70 - 10Q = 10 \Rightarrow Q_m = 6, \quad P_m = 40, \quad \pi = 240 - 220 = 20$$

$$DWL = \frac{1}{2}(40 - 10)(12 - 6) = \mathbf{90} \quad (\text{efficient: } 70 - 5Q = 10 \Rightarrow Q^* = 12)$$

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Working through it

The firm. From Part 1: $P = 70 - 5Q$, $TC = 160 + 10Q$, with Q in thousands of homes.

The signature of natural monopoly. $MC = 10$ and $ATC = 160/Q + 10$, so **marginal cost sits below average cost at every quantity**. That is not an accident of these numbers — it follows from a large fixed cost and a small constant marginal cost, and it is what makes the regulatory problem insoluble.

Unregulated behavior. It is an ordinary monopolist. MR falls twice as fast as demand: $MR = 70 - 10Q$. Setting $MR = MC$:

$$70 - 10Q = 10 \implies Q_m = 6, \quad P_m = 40$$

Profit: $240 - (160 + 60) = \mathbf{20}$.

The damage. The efficient quantity solves $70 - 5Q = 10$, giving $Q^* = 12$. So

$$DWL = \frac{1}{2}(40 - 10)(12 - 6) = \mathbf{90}$$

Half the city's homes go unserved, and the loss is \$90 thousand a month. That is the disease the next two frames try to treat.

Attempt 1: cap the price at marginal cost

Attempt 1: cap the price at marginal cost

Set $P = MC = 10$. Buyers take $70 - 5Q = 10 \Rightarrow Q = 12$ — the efficient quantity.
 $DWL = 0$. **Perfect?** Check the firm's books:

revenue	10×12	120
variable cost	10×12	120
fixed cost (the wires)		160
profit		-160

At $P = MC$, revenue covers the variable cost **exactly** — and not one dollar of the wires. **The loss is the entire fixed cost.** The firm exits; the regulator has efficiently priced the utility out of existence.

Working through it

The intervention. Set $P = MC = 10$ — the textbook efficiency prescription.

What it achieves. Buyers take $70 - 5Q = 10 \Rightarrow Q = 12$, the efficient quantity. Every home worth connecting is connected, and **deadweight loss is zero**. On the welfare criterion this is a complete success.

Now check the firm's books.

- Revenue: $10 \times 12 = 120$.
- Variable cost: $10 \times 12 = 120$.
- Fixed cost (the wires): 160.

$$\pi = 120 - 120 - 160 = -\mathbf{160}$$

Why the loss is exactly the fixed cost, and this is general. With constant marginal cost, pricing at marginal cost means every unit's revenue exactly equals its variable cost. Revenue therefore covers the variable bill and not one dollar of the fixed bill — so the loss is the whole fixed cost, whatever the demand curve happens to be.

The consequence. The firm exits. The regulator has efficiently priced the utility out of existence, and the city has no electricity distribution at all.

Attempt 2: cap the price at average cost

Attempt 2: cap the price at average cost

Let the firm break even: set $P = ATC$ and let the equation pick the point.

$$70 - 5Q = \frac{160}{Q} + 10 \Rightarrow 5Q^2 - 60Q + 160 = 0 \Rightarrow (Q - 4)(Q - 8) = 0$$

Take the larger root (more homes, lower price): **Q = 8, P = 30**.

Check: $ATC = 160/8 + 10 = 30 = P$ — break even, exactly. The firm survives without a subsidy.

But $P = 30 > MC = 10$: homes 8,000–12,000 still go unserved.

$$DWL = \frac{1}{2}(30 - 10)(12 - 8) = \mathbf{40}$$

Working through it

The intervention. Let the firm break even: set $P = ATC$ and let the equation locate the point.

$$70 - 5Q = \frac{160}{Q} + 10$$

Multiply through by Q and rearrange: $5Q^2 - 60Q + 160 = 0$, that is $Q^2 - 12Q + 32 = 0$, which factors as $(Q - 4)(Q - 8) = 0$.

Choosing the root. Two solutions, $Q = 4$ and $Q = 8$. Take the **larger**: more homes served at a lower price, which is the pro-consumer choice among two break-even points.

The outcome. $Q = 8$, $P = 70 - 40 = \mathbf{30}$. Verify: $ATC = 160/8 + 10 = 30 = P$. ✓ The firm survives with no subsidy.

What it costs. $P = 30$ still exceeds $MC = 10$, so homes 8,000 to 12,000 remain unserved:

$$DWL = \frac{1}{2}(30 - 10)(12 - 8) = \mathbf{40}$$

So this is a genuine improvement — deadweight loss down from 90 to 40, price down from 40 to 30, 2,000 more homes connected — but it is not efficiency, and it cannot be.

The regulator's menu — no free lunch on it

The regulator's menu — no free lunch on it

	Q (000s)	P	firm's profit	DWL
unregulated monopoly	6	40	+20	90
cap at ATC (break even)	8	30	0	40
cap at MC (efficient)	12	10	−160	0

Sliding down the menu buys efficiency **with the firm's solvency**:

$P = MC$ is efficient but bankrupts; $P = ATC$ survives but wastes. Pick your poison.

Real rate regulation lives between the last two rows — plus a quieter cost: a firm *guaranteed* break-even has little reason to cut costs.

Working through it

The three rows, read together.

- **Unregulated:** $Q = 6$, $P = 40$, profit $+20$, $DWL = 90$.
- **Cap at ATC :** $Q = 8$, $P = 30$, profit 0 , $DWL = 40$.
- **Cap at MC :** $Q = 12$, $P = 10$, profit -160 , $DWL = 0$.

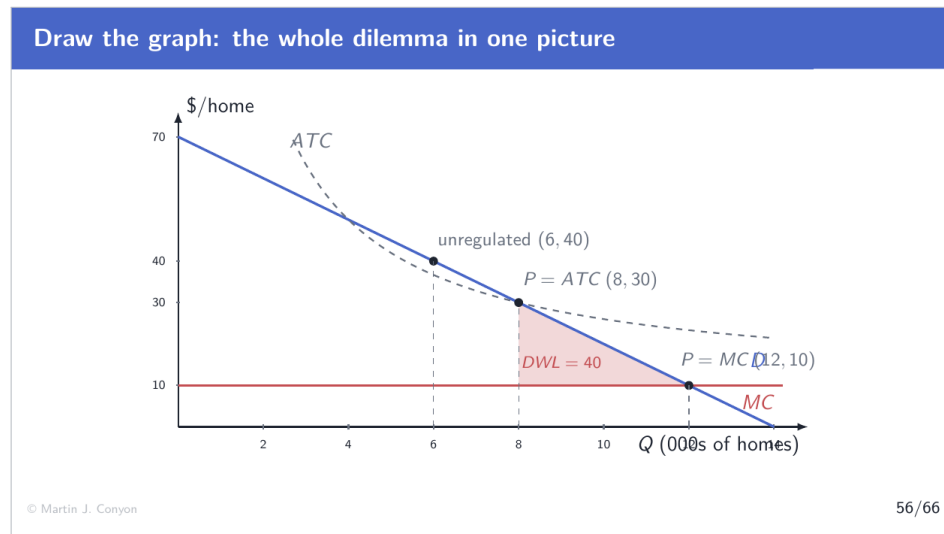
The pattern. Sliding down the menu buys efficiency **with the firm's solvency**. Every step that reduces deadweight loss also reduces profit, and the last step takes it below zero.

Why no fourth row solves it. The problem is structural: $ATC > MC$ everywhere for a natural monopoly, so a price equal to MC is necessarily below ATC , and a price that covers ATC is necessarily above MC . There is no price that is both efficient and solvent. Only a subsidy escapes the bind, and a subsidy has to be funded by taxes that have their own deadweight loss.

Where real regulation lives. Between the last two rows — some form of average-cost or rate-of-return regulation.

And a quieter cost of it. A firm guaranteed break-even has little reason to cut costs, because savings are regulated away rather than kept. Rate regulation controls the price and blunts the incentive to be efficient.

Draw the graph: the whole dilemma in one picture



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Working through it

What the figure shows. Demand, marginal revenue, a marginal cost line at 10, and an average cost curve sloping down across the whole diagram and lying *above* MC everywhere.

Why ATC lies above MC throughout. ATC includes a share of the fixed cost and MC does not, so the two can only meet if the fixed cost is spread over infinitely many units. For a natural monopoly the gap never closes.

Reading the three policies as three heights.

- The **unregulated** price sits high on the demand curve, above ATC : the firm profits.
- The **average-cost** price is where demand meets ATC : the firm exactly breaks even.
- The **marginal-cost** price is where demand meets MC : it lies *below* ATC , so the firm loses.

The dilemma made visual. Because ATC is everywhere above MC , the break-even point and the efficient point are different points on the demand curve — always, in every natural monopoly. That single geometric fact is why the menu has no good row on it.

Options 3 and 4: own it, or leave it alone

Options 3 and 4: own it, or leave it alone

Public ownership. The city runs the grid itself and can price at MC , eating the loss openly through taxes.

- The catch: no owner hunting for cost cuts — losses land on taxpayers, and padding survives.

Do nothing. If DWL is small and regulators are imperfect, tolerating the monopoly can be the least-bad option.

- Stigler's warning: "political failure" from imperfect regulation can cost more than the "market failure" it treats.

Four options, four flaws. Policy picks the cheapest flaw, not a free fix.

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Working through it

Public ownership. The city runs the grid itself and can price at MC , covering the resulting loss openly through taxation. This is a real solution to the pricing dilemma — the efficient price becomes affordable because the shortfall is funded rather than borne by an owner.

The catch. Nobody is hunting for cost cuts. A private owner keeps what it saves; a public enterprise's losses land on taxpayers, so padding survives. The efficiency gained on price can be lost on cost.

Do nothing. If the deadweight loss is small and regulators are imperfect — badly informed, slow, or captured by the industry — tolerating the monopoly can be the least-bad option.

Stigler's warning. George Stigler, who won the Nobel Prize for studying regulation, argued that "political failure" from imperfect regulation can cost more than the "market failure" it treats. Regulators depend on the firm for the cost data they regulate with, which is not a promising arrangement.

The honest summary. Four options, four flaws. Policy picks the cheapest flaw, not a free fix — and which is cheapest depends on facts about the particular industry and the particular regulator.

A third case: monopolistic competition

A third case: monopolistic competition

A monopoly has **one** seller and **barriers** that keep it alone. Perfect competition has **many** sellers of an **identical** product.

Between them sits a third case, and it is where a great deal of what you actually buy comes from: **many sellers, each offering a slightly different product**.

- Because the products differ, each firm faces its *own* downward-sloping demand curve. In the short run it behaves exactly like the monopolist we have just studied: it sets $MR = MC$ and reads the price off its demand curve.
- **But entry is free.** Nothing keeps rivals out.

That one difference changes the long run completely. A monopoly keeps its profit because entry is blocked. Here, profit attracts entrants, each takes customers away, and the profit is competed to nothing.

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Working through it

Monopoly and perfect competition are the two poles. A monopoly has one seller, protected by barriers. Perfect competition has many sellers of an identical product, and no firm has any influence over the price.

Between them lies monopolistic competition: many sellers, each offering a product slightly different from the others. Restaurants, hairdressers, brands of coffee, running shoes.

Two features matter, and they pull in opposite directions.

Because the products are differentiated, each firm has some pricing power. Its demand curve slopes down, so it faces exactly the problem set out above: marginal revenue lies below price, the firm sets marginal revenue equal to marginal cost, and reads the price off the demand curve. Every tool built here carries straight over.

Because entry is free, that pricing power cannot last. In a monopoly the barriers are what preserve the profit. Take the barriers away and profit becomes a signal: it attracts entrants, each entrant takes some of the incumbent's customers, and the incumbent's demand curve shifts left. Entry stops only when there is nothing left to attract anyone.

So the short run is the machinery built here, applied unchanged. What differs is the long run, and that is why monopolistic competition is treated as a model in its own right rather

than as a footnote to this one.

7. Part 7: Extensions: The Austrian Critique

Competition as a process, not a state

Competition as a process, not a state

This model is a **snapshot**: one market, one moment, deadweight loss measured against a frozen benchmark.

Hayek (1948) and Kirzner (1973): that confuses competition's *outcome* with competition itself —

- Competition is a **process of discovery**: entrepreneurs hunting profit, innovating, adjusting.
- A market with one seller may still be disciplined — **by the threat of entry**.

Baumol (1982) formalized it: in a **contestable** market (costless entry *and* exit), even a lone incumbent prices near cost. *What matters is the door, not the head-count.*

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Working through it

Everything in this part is beyond the department's required material. It is included because a student who can compute a deadweight loss triangle should also know what serious economists say against the exercise.

The charge. The model just built is a **snapshot**: one market, one moment, a loss measured against a frozen benchmark. Hayek (1948) and Kirzner (1973) argue that this confuses competition's *outcome* with competition itself.

Their alternative. Competition is a **process of discovery** — entrepreneurs hunting profit, trying things, innovating, adjusting. On this view the interesting question is not whether a market currently satisfies $P = MC$ but whether the process of discovery is operating.

The sharpest version of the point. A market with one seller may still be disciplined — **by the threat of entry**. Baumol (1982) formalized this: in a *contestable* market, where entry and exit are both costless, even a lone incumbent prices near cost, because any attempt to

raise the price invites immediate entry.

So the head count may be the wrong thing to measure. What matters is the door, not the number of firms currently inside it.

Creative destruction

Creative destruction

Schumpeter (1942): the competition that matters is not price rivalry among clones — it is the **new product** that makes the old one obsolete.

- Monopoly profit is the **prize** that funds and motivates innovation — the patent bargain, economy-wide.
- Monopolists are rarely undercut; they are **leapfrogged**.

On this view, today's deadweight triangle buys tomorrow's new industries — tax it away and the engine slows.

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Working through it

Beyond the department's required material.

Schumpeter's recasting of the villain. In the analysis of Part 4, monopoly profit is a transfer extracted from consumers — \$320 thousand moved from buyers to owners. Schumpeter argues this misreads what the profit is *for*.

The alternative account. Monopoly profit is the **prize** that funds and rewards innovation. Firms invest in risky, expensive development precisely because success yields a temporary monopoly. And the innovation that wins the prize *destroys the previous monopoly* — which is why he called the process **creative destruction**.

The implication for the measurement. A snapshot showing deadweight loss may be measuring the temporary cost of a mechanism whose dynamic benefits are far larger and never appear in the triangle. The triangle counts the trades lost this year; it cannot count the product that exists at all because somebody expected to win a monopoly with it.

The honest tension. This is the same trade-off the patent frame in Part 1 described, generalized. The difference is that the patent version treats the monopoly as a deliberate policy cost; Schumpeter treats it as the engine of growth itself, which is a much stronger claim and a much harder one to measure.

The knowledge problem — and the state's own hand

The knowledge problem — and the state's own hand

Two more thrusts:

- **The counterfactual is invented.** Our $DWL = 160$ used the competitive price of a market that *does not exist*. Hayek: prices are discovered by competition — absent the process, the “efficient” benchmark cannot simply be looked up.
- **Follow the barrier.** Rothbard (1962): the durable monopolies are mostly **government-made** — patents, licenses, franchises. Private walls crumble; legal walls are maintained. The fix is repeal, not regulation.

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Working through it

Beyond the department's required material.

The first strand: what the regulator would need to know. Every policy in Part 6 required somebody to know the demand curve and the cost curves — to set a price at ATC , you must know ATC . Those numbers exist in nobody's head. They are estimated, from data the firm itself supplies, and the firm has every incentive to overstate its costs.

Why this is not merely a practical difficulty. It is the same knowledge problem that arises for central planning, applied at a smaller scale. The information needed to regulate correctly is dispersed and partly unknowable, and the regulated party controls most of it.

The second strand, and it is the more uncomfortable one. Look back at the three barriers in Part 1. Patents, licences and exclusive franchises are all *created by the state*. A considerable share of observed monopoly is manufactured by the same authority that then proposes to

cure it.

The policy conclusion the Austrians draw. Before regulating a monopoly, ask what is protecting it. If the answer is a licence, a franchise or an import restriction, the cheapest remedy is to withdraw the protection rather than to add a second layer of intervention on top of the first.

What survives the critique

What survives the critique

The critique lands real blows: the model is **static**, the benchmark **unobservable**, innovation **absent**.

But it overreaches as a blanket pardon:

- Some monopolies persist for *decades* — network effects, infrastructure, and yes, government protection.
- Creative destruction takes time; consumers pay the markup **while they wait**.

The static model is incomplete, not wrong: withheld output and killed trades are real.
Judge each monopoly with both lenses.

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Working through it

Beyond the department's required material — but the conclusion matters for how you use everything else.

What is untouched. The **arithmetic**. Given a demand curve and a cost curve, marginal revenue really does lie below price; the firm really does stop where $MR = MC$; the price really does exceed marginal cost; and the triangle really is value that nobody receives. None of that is in dispute, and none of it depends on the Austrian objections being wrong.

What is disputed. Not the calculation but its **standing as a verdict**. Three specific claims:

- The benchmark is *static*, and competition may be better understood as a process than as a state.

- The profits may be doing dynamic work — funding innovation — that the snapshot cannot see.
- The proposed cure has failure modes of its own, and is sometimes aimed at a barrier the state itself erected.

How to hold both. Use the model to compute what a monopoly costs at a moment — it does that well and nothing else does it better. Then be careful about moving from that number to a policy recommendation, because the number answers a narrower question than the recommendation requires.